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Digital age crafting from misfit wood: A bundled pillar

Jaakko Torvinen
Department of Architecture, Aalto University, Finland

Jakob Sieder-Semlitsch, Jens Pedersen, Anders Kruse Aagaard & Niels Martin Larsen
Aarhus School of Architecture, Denmark

Matti Kuittinen
Department of Architecture, Aalto University, Finland

ABSTRACT: This study investigates the possibilities of leveraging digital tools and machining processes to construct structural components from misfit wood – harvested wood considered unsuitable for the wood product industry due to its dimensional or quality characteristics. This research learns from an intuitive robotic instruction method, where drawings made directly on the work piece are used to instruct robotic action. It exhibits the method through an architectural demonstrator made of three irregularly shaped whole logs, aligning with a broader theme of repurposing overlooked materials against a backdrop of increased environmental consciousness in the architectural domain. The feasibility of this approach is sustained by an underlying hypothesis that posits the possibility of manufacturing with naturally shaped round wood without traditional crafting skills, guided by hand-drawn marks that inform the desired machining operations.

1 INTRODUCTION

1.1 *Utilization of irregular forest materials in traditional and contemporary architecture*

We are on the verge of a climate disaster (Ripple et al., 2024), and the construction sector has been identified as having the potential to reduce greenhouse gas emissions by 66% (IPCC, 2023), through, for example, increased use of biobased materials such as wood (Buchanan & Levine, 1999; Churkina et al., 2020; Eriksson et al., 2012). However, current architectural processes are focused on standardized straight elements, which results in large waste streams from the forest industry, where valuable curved wood, termed misfit wood, is often overlooked. In the context of this study, *misfit wood* refers to harvested wood that is considered unsuitable for utilization in the wood product industry due to its dimensional or quality characteristics, such as being excessively thick or thin, lacking straightness, or free from visual defects. In Finland, for example, less than a third of felled tree trunks are harvested for sawlogs, while the rest are used for pulp or energy production, which is where more than half of the sawlog residues ends up during processing (Kulju et al., 2023). To challenge this traditional paradigm, the current study examines the role of traditional building craftsmanship in which forest resources, specifically irregularly shaped wood, directly inform architectural design rather than being discarded in favor of standardization (Zwerger, 2023, pp. 10–32). The evidence of this craft is notable in various examples. For instance, it can be seen in the use of root knees in Norwegian stave churches (Zwerger, 2023), the construction of a pedestrian bridge utilizing living aerial roots (Watson, 2020), and in Danish tobacco barns, where irregular wood is incorporated instead of straight members (Arkiv.dk, n.d.). Moreover, parallels can be drawn with the ancient Egyptian practice of utilizing bundled wood poles or reeds to form sturdy pillars supporting roof structures (Smith, 1938, pp. 249, 254–255), as well

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as the method of using reed bundles in mudhif houses in the wetlands of Iraq (Watson, 2020). This architectural expression is also evident in modern architecture, such as the design of Alvar Aalto's Villa Mairea in 1939, where bundling was employed as a means of architectural expression (Dietziker & Gruntz, 2022).

Digital fabrication technologies have previously been shown to enable the processing of misfit wood, but often through a complex scanning process (Bukauskas et al., 2017; Larsen et al., 2022; Mollica & Self, 2016). This study explores the potential of an intuitive method for utilizing the fabrication process, which is demonstrated through a process of manufacturing a building element made of three irregularly shaped whole logs, drawing inspiration from the precedents (Figure 1).

1.2 *State of the art*

Current research on whole wood has focused on the structural strength (Moreno Gata et al., 2024; Ranta-Maunus, 2000; Wolfe, 2000) and matching the specific geometry of irregular wood, e.g., branch joints or curved timber, to a certain designed structure where the material used in the design is selected to fit a given geometric constraint (Bukauskas et al., 2019; Larsen et al., 2022; Mollica & Self, 2016). A commonality among these examples is the complexity associated with machining misfit wood using digital fabrication technologies. The accurate positioning of the physical member and the associated computer programs for machining necessitates a precise digital model of each element to be machined. Previously, this complexity has been mitigated through multiple digital processes, such as 3D scanning, post-processing, and geometric rationalization, as demonstrated by Larsen et al. (2022), Mollica & Self (2016), and Vestartas et al. (2020).

Instead of following this modus operandi, this research learns from an intuitive robotic instruction method, where drawings made on the work piece are used to instruct robotic action, as previously demonstrated by Pedersen et al. (2020, 2021). But instead of using a regular camera to see the drawing, this research employs a stereo camera, whereby an accurate representation of the drawn and geometric complexities of misfit wood can be achieved.

1.3 *Objectives and scope of the study*

The objective of the research was to investigate the potential of novel digital tools and machine tools in conjunction with a previously untapped source of material: misfit wood. To achieve this, a prototype was developed using wood sourced from a local sawmill's waste piles as the primary building material. This investigation delves into the digital design and machining stages of the prototype, emphasizing the unique challenges posed by working with naturally shaped round wood in the context of digital manufacturing.

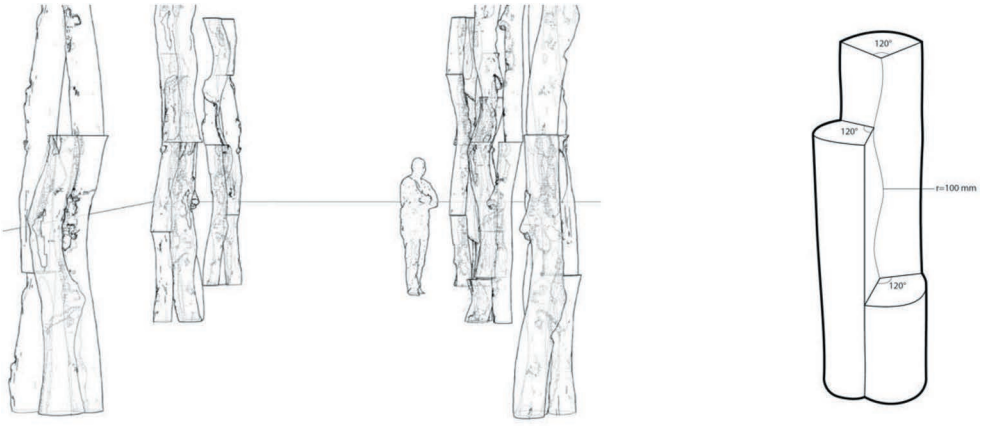


Figure 1. The bundled pillar of three irregular shape logs is connected to each other with sawn surfaces. Connecting pieces laterally and vertically by stacking them overlapping, the element can be increased in two directions. The two sides of each log, at an angle of 120 degrees to each other, are sawn into a wavy pattern with a radius of 100 mm.

The prototype employs a joint and workflow system that accommodates these unique shapes without necessitating individual adjustments (Figure 2). The present study does not address the material or structural properties, which the conclusion suggests should be the subject of further research. In making the prototype, the logs were machined without prior scanning and visually positioned by the operator.

The underlying hypothesis posits that it is feasible to construct with naturally shaped round wood without traditional craft skills by utilizing digital tools and machining processes guided by hand-drawn marks that inform the desired machining operations.

2 MATERIALS AND METHODS

2.1 *Iterative design-led approach for manufacturing architectural components*

This research introduces a method for manufacturing architectural components from misfit wood using an iterative design-led research approach. The development of the method is evaluated and documented through notes, photographs and videos of the process, and is detailed in Figure 2, which shows the transition from salvaged wood to a machined architectural wood element.

The research took place in spring 2024 at the Aarhus School of Architecture and utilized an ABB IRB 6620 robot equipped with a stereoscopic depth sensor: Real Sense, and a bandsaw with a 10 mm wide saw blade. Prior to the construction of the 1:1 demonstrator, the research team initiated a series of tests to assess the robot's manufacturing capabilities. These tests involved the implementation of diverse cutting strategies and the simultaneous assessment of the stereoscopic depth sensor.

To mitigate the complexities described in section 1.2, this research builds on top of a known intuitive method for instructing robotic actions; here, a camera and computer vision algorithms enable the extraction of fabrication information from physical drawings made directly pieces (Pedersen et al., 2020, 2021). In its current form, the method allows for the processing of flat workpieces, which is dissimilar to the complex three-dimensional shapes of round misfit wood. Instead, this study used a stereoscopic depth sensor, Real Sense, rather than a calibrated web camera as in the previously mentioned studies. The Real Sense camera produces both photos and colored point clouds, which allowed for the adoption of methods from existing research, whereby drawings could be extracted from the color information of the point cloud. The Real Sense was mounted on the robot (Figure 3) and navigated to a detection pose when a log was positioned in the robot cell.

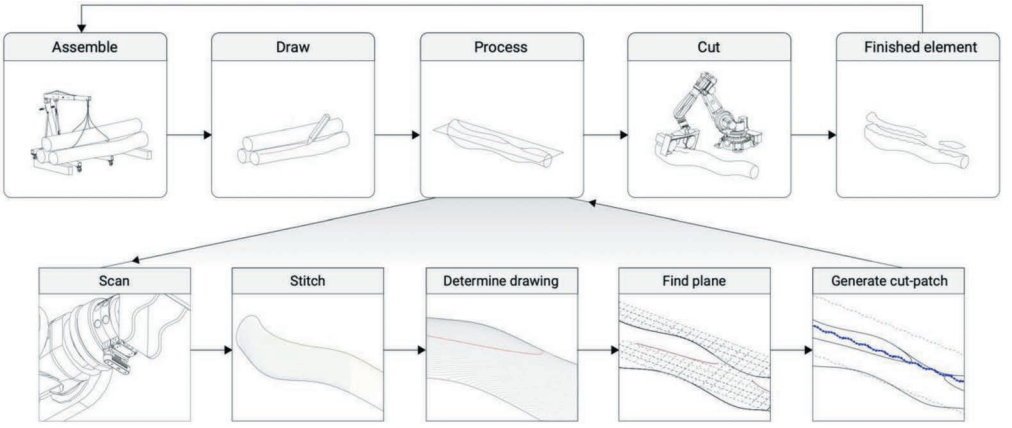


Figure 2. The prototype employs a joint and workflow system that can accommodate unique shapes without necessitating individual adjustments. Three logs were assembled to the anticipated orientation. A straight line was drawn on the workpiece and marked with a red marker. The log was then scanned, from which the red mark was extracted and used to guide the machining process. This created a machining path for the robot. The process was repeated for all three irregularly shaped logs.

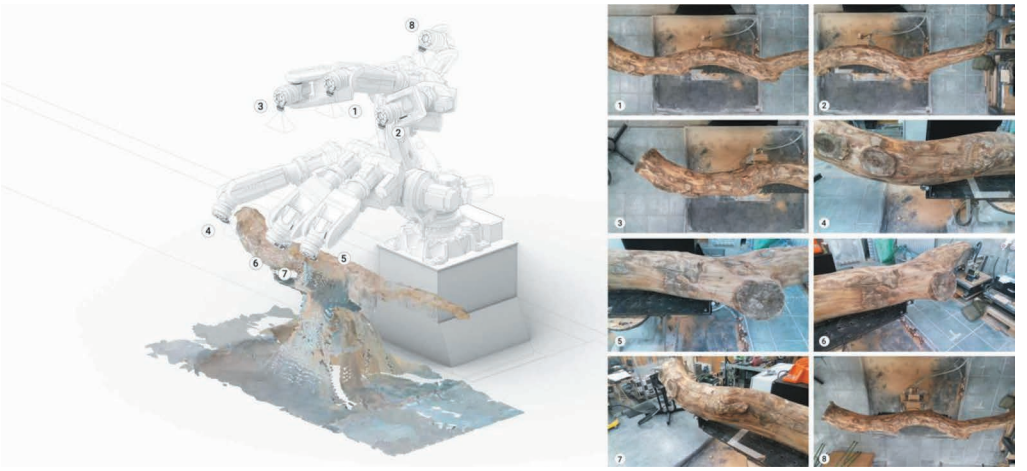


Figure 3. The Real Sense camera produces both photos and colored point clouds. The Real Sense was mounted on the robot and navigated to a detection pose, when a log was positioned in the robot cell.

From the detection pose, it was possible to identify the position of logs within 10 mm through a digital method that, due to the limited scope of this paper will be presented in a technical paper at a later stage.

2.2 *Robotic setup and fabrication limitations*

The different cutting strategies that were explored entailed curvature and double curvature cuts, ruled cuts, spiral cuts, and radial cuts. From these initial tests it was found that tool jamming would occur during both spiral and double curved cuts, and if the cutting radius was below 100 mm. Therefore, the research continued with tests that utilized a 100 mm cutting radius as optimal.

Following this finding the research performed a scaled fabrication test for a bundle pillar section made from three 10-centimetre diameter round wood. These logs were sawed with two wavy faces at 120 degrees to each other (Figure 1), but for the third piece, it was found that the wavy sawing had to overlap to enable the pieces to join freely.

In addition to the aforementioned objectives, the Real Sense camera and its potential as a tool were also evaluated during the test phase. The results demonstrated that this approach proved to be an effective and expedient method for conducting tasks, and it was subsequently integrated into the prototyping process.

2.3 *Workflow and machining strategy*

The material was sourced in January 2024 from a Danish sawmill, where wood that could not be used in the production line had been discarded. Three beech logs with diameters ranging from 200 to 230 mm and lengths between 2.8 and 3.2 m, were salvaged from these.

The logs were debarked using a jet washer and subsequently placed in an indoor environment for a period of two months for drying. It was of the utmost importance to remove the bark to expedite the drying process and prevent the infestation of insects that would otherwise feed on the sapwood.

The logs were stacked horizontally and a guide line was delineated on them with a red marker using a laser as a reference line (Figure 2 & Figure 4), indicating the anticipated point of contact between the logs after the completion of the machining process. Subsequently, the first log to be machined was positioned on a rotating table and fixed within the operational zone of the robot, enabling the detection and cutting of two drawings (Figure 2 & Figure 4). The log was scanned with a stereoscopic depth sensor mounted on the robot. The images and

color point cloud, both produced by the sensor, were exported to the 3D modeling software Rhinoceros 3D, where the position of the piece in relation to the robot was verified. From the same visual data, a red line was extracted and superimposed on the surface of the log to be machined. The line was then used for the automatic development of the machining strategy in Rhinoceros 3D and the visual programming environment Grasshopper3D. The logs were fixed in sequence to the robot's operational zone and scanned with the Real Sense technology, and the red reference marks on the logs were readily discernible. The processing of logs was conducted in accordance with the following steps:

1. The piece was fixed to the work area.
2. The Real Sense was used for scanning.
3. The development program was initiated.
4. The program was executed.
5. The piece was removed from the work area.

Once the three logs had been machined, they were stacked together again horizontally, with the wavy surfaces aligned. The wavy surfaces (Figure 5) fit together well, which was anticipated since the machining process involved the repetition of the same machining pattern. Subsequently, the logs were fastened together with the aid of cargo straps. Thereafter, three 12 mm holes were drilled in the upper extremity of the bundled pillar, with two logs being traversed with each hole. Ten-millimeter threaded rods were inserted into the apertures and secured with washers and nuts. Subsequently, the bundled pillar was mounted on the robot's machining area. A transverse line was drawn at both ends of the bundled pillar with the aid of a laser, ensuring that it was perpendicular to the center line of the pillar. The line was identified by a Real Sense scan, which was then used to execute the robot's sawing program. A chainsaw was mounted on the robot and used to saw both ends of the bundled column. Following this, a rectangular flat base made of sheet metal was fixed to the lower end of the bundled column with structural screws.

3 RESULTS

3.1 *Efficiency and versatility of workflow and machining process*

The preliminary trials and the production of the final bundle pillar's three constituent elements effectively established an efficient production process. As detailed in Section 2.3, the workflow and machining strategy permitted the machining of irregular round woods using digital software and machine tools, including a stereoscopic depth sensor. The RealSense technology was employed in conjunction with a six-axis robot, equipped with an array of woodworking tools.

The machining method demonstrated versatility, successfully handling round workpieces of various irregular shapes in multiple directions. The system facilitated a direct chain of commands

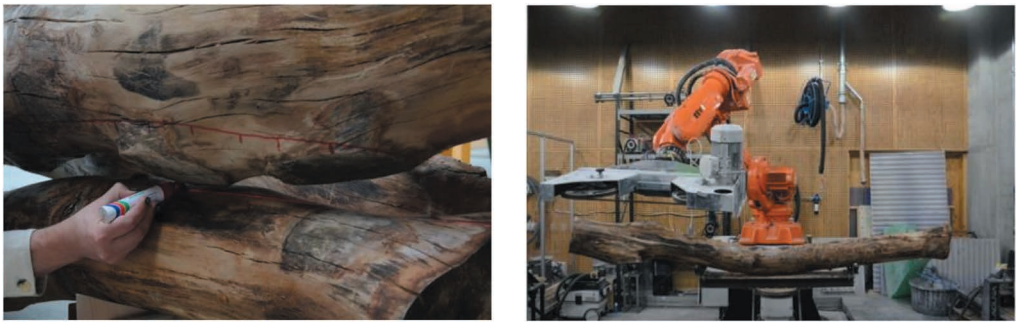


Figure 4. The drawn line was used as an alignment line – red with the stereoscopic depth sensor: Real Sense – to identify the location of the first machining operation.




Figure 5. Horizontal lines were drawn on the logs in red marker to show where the log would be cut. Small cross lines indicate which side is the side to be removed. Each log was placed on a workbench and 3D scanned using the same 6-axis robot used in the machining process. This allowed the log to be positioned freely.

between the hand-drawn markings and the machine tool, thereby expediting the scanning phase and subsequent identification of the position and shape of the object set for machining. Prior to the scanning and machining processes, the workpiece could be freely positioned within the designated machining area, allowing for adjustments and repositioning during the manufacturing process.

3.2 *Overview of the final prototype*

The final prototype (Figure 6), a bundled column crafted from three salvaged misfit wood logs, has a height of 2285mm with the diameter at its widest point being 550mm. Each curved and irregularly shaped logs contributes to the pillar's distinctive, rectilinear form. The prototype illustrated the aesthetic contrast between the rough surfaces of the logs and the machined cuts. The visual characteristics of wavy sawing surfaces are particularly evident in instances where the logs are not in contact with one another, due to their inherent curvature. Some of the hand-drawn markings, which guided the machining process, are still visible on the pillar's surface, serving as a testament to the digital workflow employed.

The prototype was exhibited at the New European Bauhaus 2024 festival in Brussels and the United Nations Climate Change Conference 2024 in Baku, Azerbaijan, garnering interest for its unique utilization of salvaged materials and exploration of digital fabrication techniques.

4 DISCUSSION

4.1 *The findings and learnings*

In the preliminary tests described in chapter 2.2, the approaches that proved effective involved accurately identifying the object using a stereoscopic depth sensor, Real Sense, and leveraging the resulting data to position the tool according to the position of the work piece and the marking made to it and initiate the cut. This method ensured precise alignment and execution. Through these findings and learnings, a suitable method to implement a 1:1 prototype was found.

The prototype successfully exemplifies the possibilities of utilizing currently untapped wood material sources with novel digital technologies and digital machining tools. Furthermore, it illustrates that it is feasible to create a construction component using three randomly selected, irregularly shaped logs. However, several obstacles arose during the implementation that would need addressing for industrial-scale production. The most significant challenge pertains to the situation created by traditional prototyping, the workflow at this stage being characterized by hand-on prototyping taking place in a fully customized situation involving specific laboratory conditions and skilled technicians. Such processes can also be conducted on an industrial scale by developing an automated production line. A further discussion could be

whether mass production is a necessary component of the industry or whether there is potential for greater mass customization in the future, utilizing existing technologies. Additionally, the handling of heavy logs and the reach limitations of a six-axis robot present their own challenges. Both issues can be addressed with industrial-sized machinery.

Overall, the prototyping phase has provided valuable insights into the practical challenges of working with irregular wood shapes and the importance of automation in enhancing efficiency and design flexibility.



Figure 6. The final prototype, a bundled column crafted from three salvaged misfit wood logs.

4.2 *Applications in architecture and construction*

This research promotes the utilization of wood in architectural design, particularly in a manner that accentuates the natural forms and patterns into what could be denoted as a brutalist aesthetic. This design approach emphasizes the inherent characteristics of wood and its organic qualities, potentially leading to an increase in wood architecture projects that seamlessly integrate with their surrounding environments. Elements made with misfit wood, like the prototype in this study, will be unique each time, depending on the natural aesthetic properties of the wood. Further development of the prototype could include structural tests and an investigation of the scalability of the design in terms of both column diameter and length. As illustrated in Figure 1, the concept of assembling misfit wood members in a stepped configuration represents a possible approach to vertically scaling the element. Further development of the prototype could also include an investigation of the optimal ratio of saw cut to material removal.

This study provides insights into longer-term applications of non-standard wood, offering an alternative to its usual short-term uses, such as pulp or energy production. The constructed prototype presents a potential solution for using misfit wood in construction, which could contribute to the goal of reducing greenhouse gas emissions in the building sector.

4.3 *Conclusions*

This study proposes the potential for innovative digital tools and workflows to resurrect the historical use of irregular, naturally shaped wood in architectural applications, thereby challenging the established norms of standardized procedures.

The development of a prototype comprising irregular whole logs exemplifies the distinctive material and aesthetic characteristics of misfit wood, thereby proffering an alternative to the prevailing tenets of contemporary construction practices.

Further research could investigate the necessity of mass production and the potential for new digital tools to facilitate mass customization. In future research, it will be essential to evaluate the potential environmental impacts of utilizing misfit wood on biodiversity, habitat, and carbon sequestration.

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